LAMMPS official dem01-hertz-tsuji

Normal collision: restitution and momentum | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Total x momentum	0 absolute limit 1.414e-05	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS
Restitution	0.5 relative limit 0.02	0.4998587 err 0.0002826 PASS	0.4998587 err 0.0002826 PASS	0.4999018 err 0.0001963 PASS	0.4999018 err 0.0001963 PASS

Check time=4.1e-05 s; original step 4100. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
left	R=0.0025 rho=2700	(-0.00255, 0, 0)	(2, 0, 0)	(0, 0, 22)
right	R=0.0025 rho=2700	(0.00255, 0, 0)	(-2, 0, 0)	(0, 0, -22)

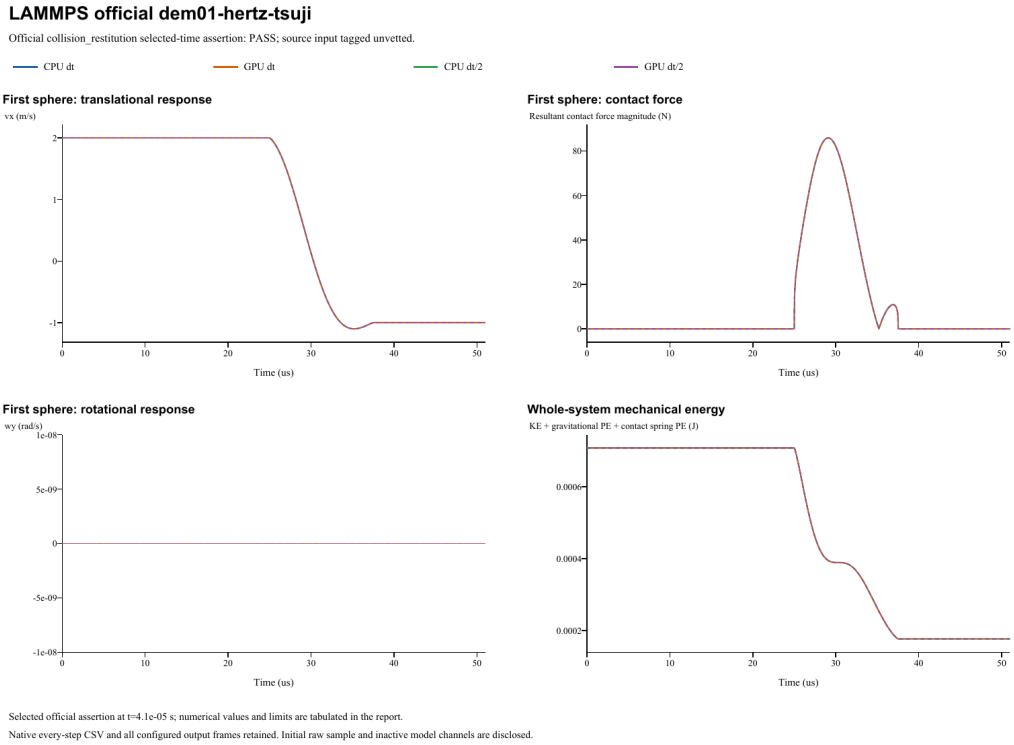
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=3.927729772e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0.4. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-08/5e-09 s; 5100/10200 steps. Exact Tsuji coefficient mapping: native beta=sqrt(2/5)*P6(e_source); native material restitution is the inverse beta transform, not a fitted rebound. Official analytic target remains e_source. Normal force prefactor and damping at every overlap match. Equal opposite pair spins or zero wall spin preserve exactly zero contact tangential velocity. Nonzero friction is retained but no active tangential damping equivalence is claimed.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob e908cea7546be4d6c0579269aef5a782b4104bd1. Official tag: unvetted. LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Total x momentum	0	0	absolute: 1.414e-05
Restitution	0.00028261	0.00019631	relative: 0.02

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 2.29e-16 m/s / 0 rad/s / 1.42e-14 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 1.19e-18 kg m/s / 2.17e-19 m / 2.14e-16 m/s. Gravity is added separately to saved native contact forces.

Complete results and verified scope

All 30,604 native frames are retained; main frame counts=[5101, 5101, 10201, 10201]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.